

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Gravity and Magnetic Methods of Prospecting (Lab)  
L-T-P: 0-0-3 Credits:2

3. Pre-requisites: NA

4. Syllabus and reference books:

### Syllabus:

Reductions of field gravity data, Application of data enhancement techniques, gravity data processing, modeling of gravity effects of simple geometrical shapes

Corrections to Magnetic data, Magnetic data processing, modeling of simple geometrical shapes

Computation of gravity and magnetic anomalies for two and three-dimensional regular shaped bodies. Testing of downward continuation and second derivative operators on simulated gravity and magnetic anomalies. Interpretation of field data collected during fieldwork.

Space for entering the syllabus

### Reference Books:

1. Applied Geophysics- Telford, Geldart and Sherifi
2. Introduction Geophysical Prospecting- Dobrin, Savit
3. Gravity and Magnetic Exploration Principles Practices and applications- Hinze, Ralph and Saad
4. Applied Geophysics- Burger, Sheehan and Jones
5. Fundamentals of Geophysics- Lowrie
6. Field Geophysics by John Milsom
7. Looking into the Earth by Alan E. Musset and M. Aftab Khan
8. Physical Principles of Exploration Methods by A.E. Beck

5. Lecture-wise break-up:

| Sl. No. | Topic                                                                                                                                                           | No. of lectures |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.      | Reductions of field gravity data, Application of data enhancement techniques, gravity data processing, modeling of gravity effects of simple geometrical shapes | 10              |

|                              |                                                                                                                                                                        |           |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>2.</b>                    | Corrections to Magnetic data, Magnetic data processing, modeling of simple geometrical shapes                                                                          | <b>10</b> |
| <b>3.</b>                    | Computation of gravity and magnetic anomalies for two and three-dimensional regular shaped bodies.                                                                     | <b>10</b> |
| <b>4.</b>                    | Testing of downward continuation and second derivative operators on simulated gravity and magnetic anomalies. Interpretation of field data collected during fieldwork. | <b>10</b> |
| <b>Total number of hours</b> |                                                                                                                                                                        | <b>40</b> |